

A. M. ISMAIL

Production Engineering Student | Software Enthusiast | Industrial Automation
📍 Chennai, India ✉ ismailsim1@gmail.com ☎ 81248 14896 🌐 Portfolio 🐙 Github 🔗 LinkedIn

Professional Summary: Production Engineering student focused on the software behind automation.

I sit right between hardware and software. My engineering background gives me a solid grasp of how machines work, but my main focus is on designing the software systems that run them. I map out the overall logic, figure out how data should flow, and design how things connect. I use AI to handle the actual coding syntax, which lets me focus 100 percentage on building reliable, smart automation systems without getting slowed down by the typing.

Projects

Sri Energy Industrial Automation [Completed]:

[SRI energy](#)

Stack: ESP32, MQTT, FastAPI, Pocketbase, React, NGINX, Docker, XAMPP

- Deployed a local-first IoT overhead crane telemetry control system featuring real-time hardware orchestration.
- Containerized a React dashboard and FastAPI middleware connected to a secure self-hosted Pocketbase backend using NGINX.
- Achieved low-latency safety controls by driving multiple ESP32 microcontrollers via a local Mosquitto MQTT broker.

Ct Ski (ML-Cloud Computing) [Completed] — Computer Technology Dept

[Infrastructure Link](#)

Stack: Docker Swarm, GlusterFS, JupyterHub, FastAPI, Prometheus, Grafana, NFS

- Architected a high-availability server cluster delivering high-performance cloud environments to CS students.
- Configured distributed storage and full infrastructure metric monitoring using Prometheus and Grafana stacks.
- Programmed a slot-based booking platform powered by FastAPI automation; launched and inaugurated by the **CEO of Tekion**.

Virtual Lab for Metrology [Completed]:

[Virtual Lab](#)

Stack: React, Docker, Nginx

- Designed a comprehensive web simulator handling physical Profile Projector operations and fundamental experiments.
- Consolidated 12 decoupled web microservices into a production ecosystem using NGINX file-based reverse proxy routing.

Seven5: Attendance Tracking App [Completed]:

Stack: Flutter, Hive, Dart, Google Drive API, Google Cloud Run

- Built an offline-first tracking mobile application syncing seamlessly with cloud dashboards via Google Drive API wrappers.
- Engineered dynamic predictive algorithms allowing users to run simulation strategies on personal leave and attendance metrics.

EggShell — Visual File Stitching Engine [Completed]:

[Demo & Docs](#)

Stack: React, JavaScript, SQLite Expression Engine

- Engineered a visual data pipeline editor that maps, filters, and stitches multi-format datasets into target outputs.
- Built a declarative JSON-driven pipeline architecture enabling multi-file transformations via a single config block.
- Integrated an inline SQLite custom script utility to parse string modifications and complex data-cleaning logic.

CWM (CLI Workspace Manager) [Completed]:

[Documentation](#)

Stack: Python, Click, Rich

- Engineered an automated command-line developer environment utility handling workspace state session mapping and task logs.

Anna University App [Completed]:

[Play Store](#)

Stack: Flutter, Dart, Hive

- Contributed to frontend implementation and UI asset scaling on the official enterprise e-Governance mobile workspace.

Experience

MITONAU Motorsports (Go-Kart Team)

Dec 2024 – Present

Chassis Design & Maintenance

- Structured structural simulations and engineering parameters on racing frames designed for the TNKC and KEC championships.

Education

Madras Institute of Technology

Aug 2023 – 2027

III yr BE in Production Engineering

CGPA: 7.73/10

School Education

HSC: 545/600 (2023)

Skills

Software & Tools: MQTT, Docker, Docker Swarm, NGINX, Pocketbase, Prometheus, Grafana, JupyterHub, Git, XAMPP, GlusterFS, FireBase.

Programming: Python, FastAPI, React, Flutter, Arduino.

CAD & Analysis: SolidWorks, Creo, NX CAD, CATIA, Abaqus CAE

Languages: Tamil, English, Arabic (Reading & Writing)